# S. S. Gokula Swamy

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## Education

**B.M.S. College of Engineering** — Bengaluru, India
*B.E. in Artificial Intelligence & Machine Learning* | ***CGPA: 7.37 / 10*** *(7th Semester)* — *2023 – 2027 (Expected)*
- **Coursework:** Machine Learning, Deep Learning, Large Language Models, Time Series Analysis, Evolutionary Computing.

**Sainik School Kodagu** — Kodagu, India
*CBSE Class XII: 88.6% (2023)* | *CBSE Class X: 91.4% (2021)* — *2016 – 2023*

## Research & Publications

**GPU-Accelerated Hybrid GA for Wind Farm Layout Optimization** — IEEE AIC 2026
*Co-Author — Accepted Research Paper (github.com/Lion4467cat/wflo-hybrid-ga)* — *2026*
- **GPU Wake Vectorization:** Vectorized the Jensen wake model in PyTorch as batched $(8, 150, 150)$ CUDA tensors across 8 wind directions, screening 150-turbine layouts on an irregular $10 \times 10$ km site without Python loops.
- **Two-Fidelity Optimization:** Rescored elite layouts via NREL FLORIS (0.88 fitness gate, 10-call/generation cap) and enforced $\geq 400$ m turbine spacing with tiered repair operators; achieved **4,029.16 GWh/yr AEP** (**+2.69% / +105.7 GWh/yr** over baseline, outperforming TOPFARM and a Jensen-only GA). *Stack: PyTorch, FLORIS, PyGAD, Shapely, PyWake, TopFarm.*

## Projects

**Verified-Agent Badge** | *1st Runner-Up, UNCLONE AI Hackathon 2026 (BMSCE & DetectifAI)* — *GitHub* | *Oct 2026*
- **Real-Time Voice Deepfake Defense:** Built a live-call verification agent scoring 6-second audio windows with DetectifAI speaker verification; any 4-second segment with synthetic confidence $\geq 0.90$ places an immediate transaction hold, overriding call-level averages.
- **Zero-Dependency Audio Forensics:** Wrote a Python standard-library PCM WAV forensics module flagging silence ratio ($> 0.15$), low room-tone RMS ($< 0.004$), and clipping, plus in-browser Web Audio voice conversion to stress-test the detector. *Stack: FastAPI, Python, Web Audio API, Uvicorn, Render.*

**NutriSense** (*Capstone*) | *PyTorch, SigLIP-2, DINOv2-S, MoGe-2, SAM 2.1, FastAPI, React 19* — *GitHub*
- **3D Dietary Metrology:** Architected an 8-subsystem single-photo pipeline for PM POSHAN mid-day meals; used Google SigLIP-2 ($\alpha = 0.65$ background attenuation) to classify 115 Indian dishes and Microsoft MoGe-2 metric depth with SVD plane fitting (3.5 mm plate tare) to reconstruct food volume.
- **Mass Fusion & Compliance Audit:** Trained a SmoothL1 mass head on frozen DINOv2-S features, fused with geometric volume via inverse-variance weighting ($w_{\text{head}} \approx 0.73$, $w_{\text{geom}} \approx 0.27$), and audited IFCT 2017 nutrient bounds using a 4,000-sample Monte Carlo comonotonic engine.

**The Greenifier** | *OpenCV, Flask, NumPy, NASA POWER API, Llama 3.3 70B, Leaflet.js* — *github.com/Lion4467cat/greenifier*
- **Geospatial Computer Vision:** Built a rooftop solar estimator fetching zoom-18 satellite tiles via Web Mercator projection and segmenting usable roof area with OpenCV (HSV thresholding, Canny edges, morphological closing).
- **LLM Energy Auditing:** Combined NASA POWER irradiance data with Llama 3.3 70B (Groq) to estimate system capacity, capex (INR 65,000/kW), payback period, and $CO_2$ offset in a Flask and Leaflet.js dashboard.

## Technical Skills

- **Languages:** Python, SQL, JavaScript (ES6+), HTML5, CSS3.
- **ML & Vision:** PyTorch (CUDA), SigLIP-2, DINOv2, MoGe-2, SAM 2.1, OpenCV, NumPy, PyGAD, Monte Carlo Simulation.
- **Web & APIs:** FastAPI, Flask, React 19, Vite, Leaflet.js, Web Audio API, Uvicorn, Groq API, NASA POWER, DetectifAI.
- **Tools & Simulation:** Git, GitHub, Linux, Docker, uv, Render, FLORIS (NREL), PyWake, TopFarm, Shapely.

## Achievements, Certifications & Activities

- **Defense & Leadership:** Qualified the UPSC NDA Written Examination (2023, top ∼0.5% nationwide); earned **NCC Certificates 'A' (Grade A), 'B' (Grade B), and 'C' (Grade B)**; named **Best Volunteer**, NSS Annual Camp 2025 (selected from 100+ participants).
- **Certifications:** Python for Data Science (IBM); Green Skills and AI (Edunet Foundation).
- **Endurance & Hobbies:** Completed self-supported 79.9 km and 58.2 km endurance cycling rides; recreational marathon runner; trekking (hobby); competitive chess player.